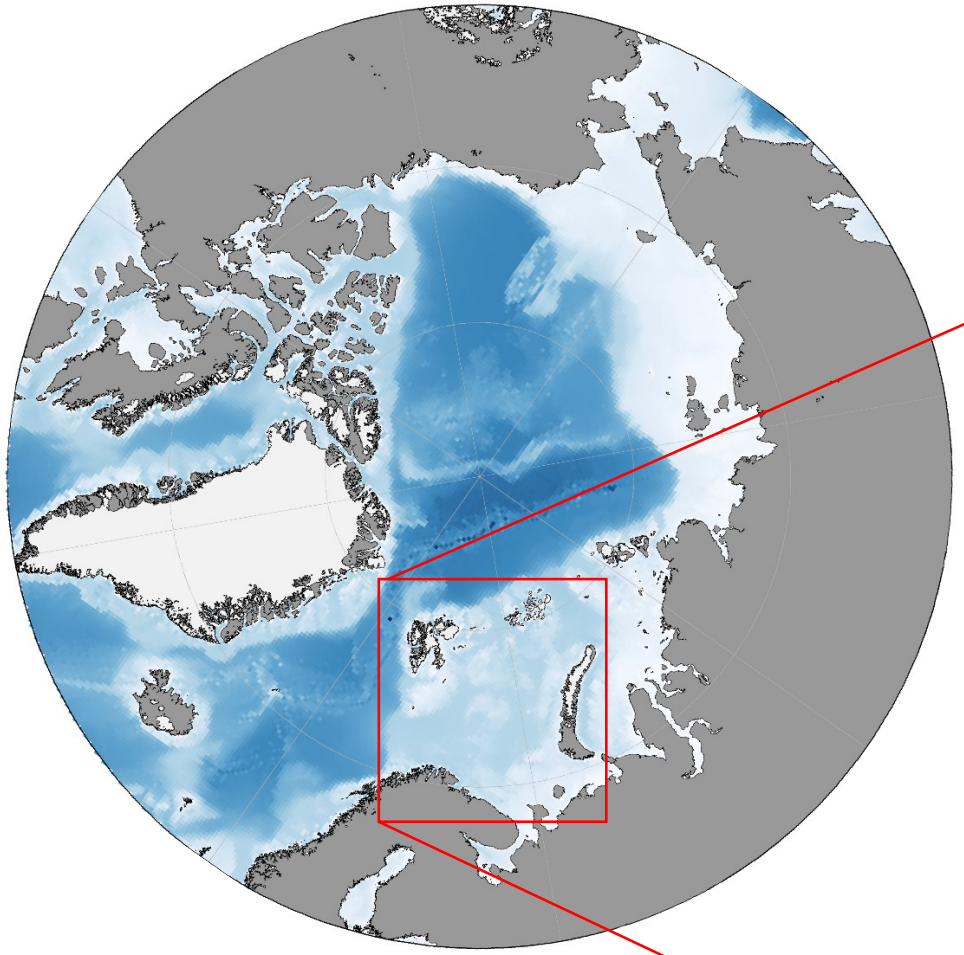
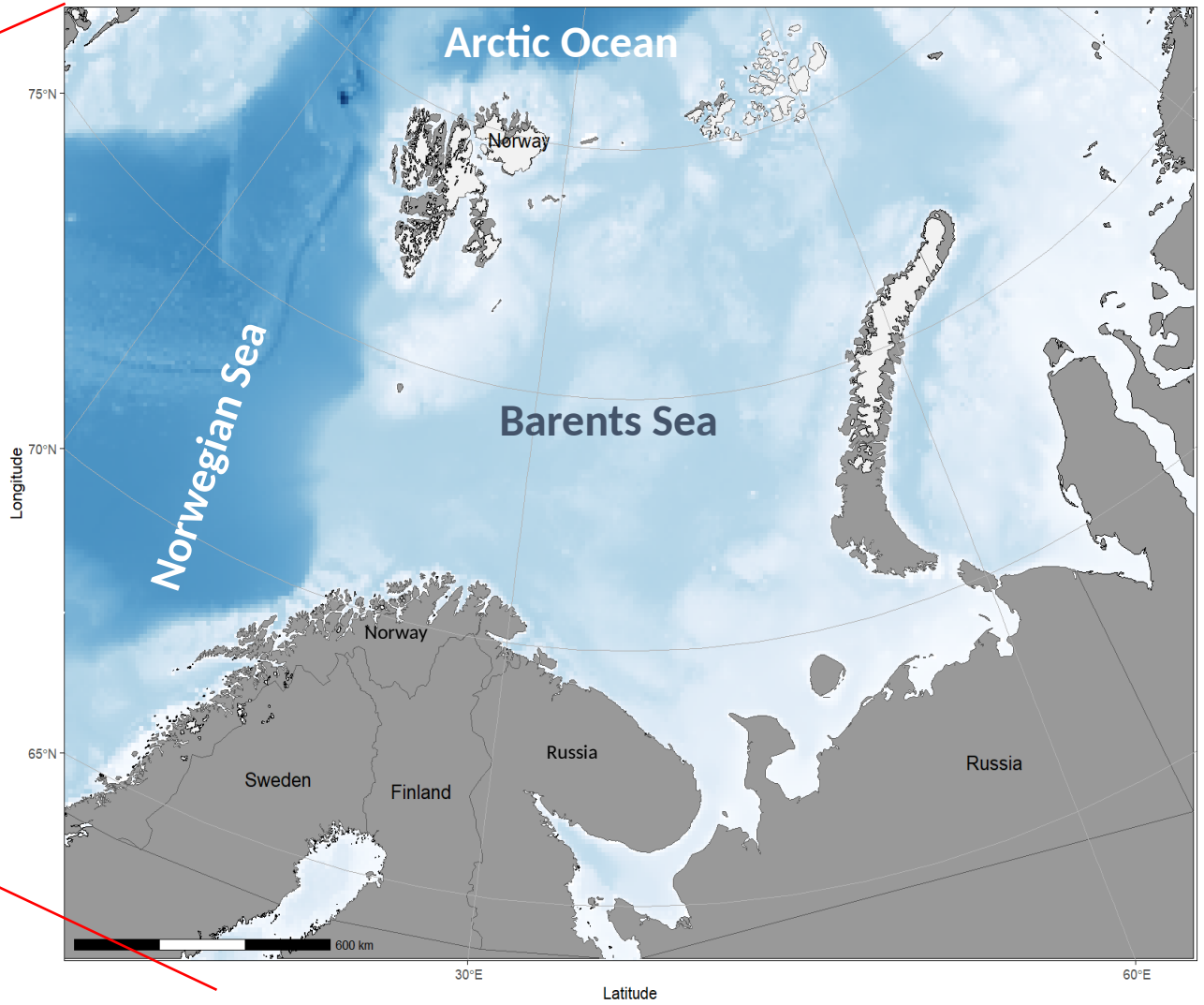
Multiple stressors in a future warmer Barents Sea: How can distributions of functional groups be affected by low and high carbon emissions scenarios?

Marcela C. Nascimento, Filippa Fransner, Robinson Hordoir, Morten Skogen, Raul Primicerio and Torstein Pedersen

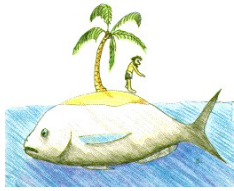
GEOMAR
Helmholtz-Zentrum für Ozeanforschung Kiel

Barents Sea




Arctic Ocean
Norway
Norwegian Sea
Barents Sea
Norway
Sweden
Finland
Russia
Russia
75°N
70°N
65°N
30°E
60°E
Longitude
Latitude
600 km

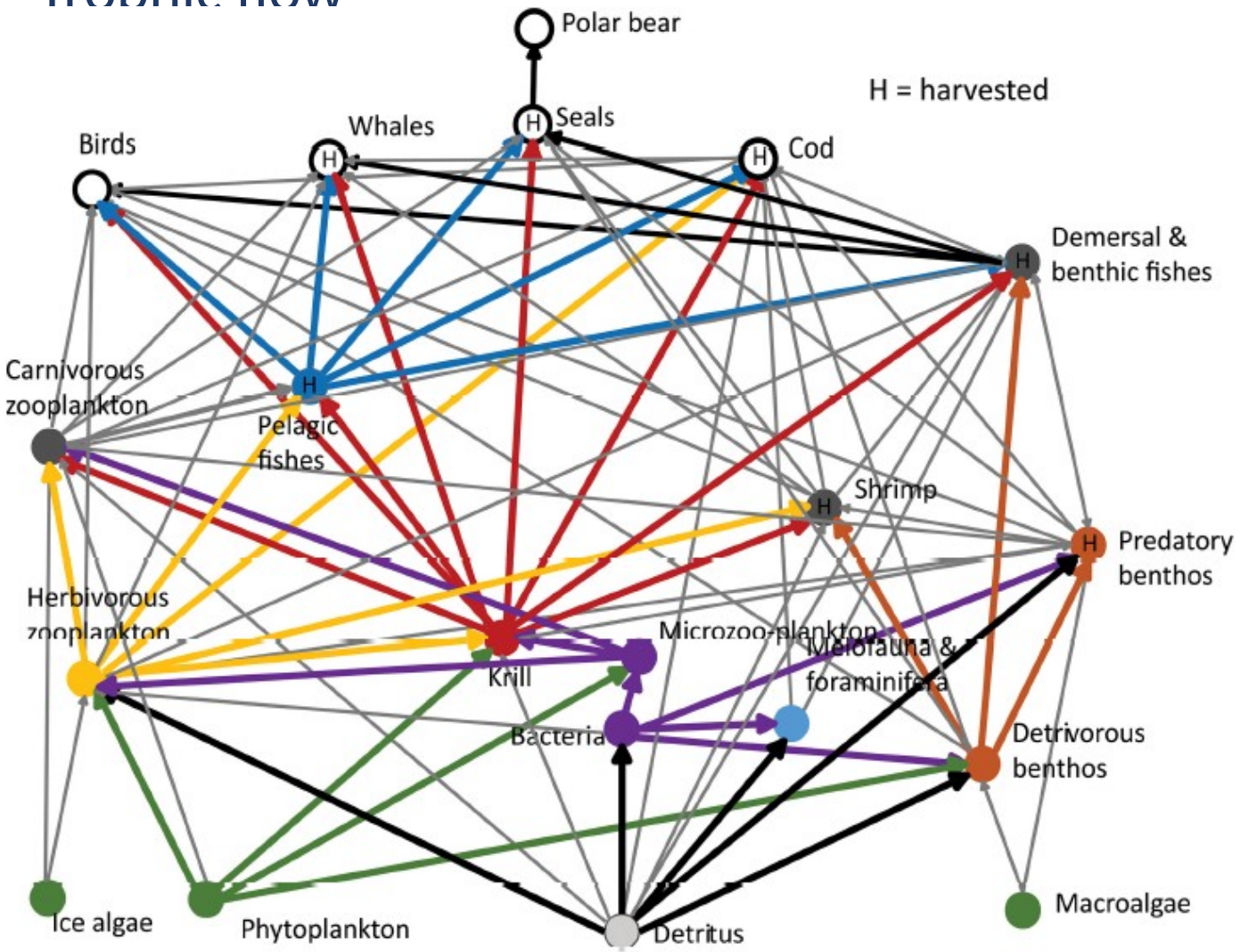
Main questions



Q1 Will the functional groups (FGs) and the community change the centres and the spread of their spatial distributions at different plausible global warming scenarios?

Q2 Are the responses different in arctic and boreal FGs?

Trophic flow



frontiers in Marine Science

ORIGINAL RESEARCH
published: 17 September 2021
doi: 10.3389/fmars.2021.732637

Overexploitation, Recovery, and Warming of the Barents Sea Ecosystem During 1950–2013

Torstein Pedersen[1], Nina Mikkelsen[1,2], Ulf Lindstrøm[1,2], Paul E. Renaud[3,4], Marcela C. Nascimento[1], Marie-Anne Blanchet[5†], Ingrid H. Ellingsen[6], Lis L. Jørgensen[2] and Hugues Blanchet[3,7]*

Contents lists available at ScienceDirect

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel

Modelling the spatial shifts of functional groups in the Barents Sea using a climate-driven spatial food web model

Marcela C. Nascimento [a,*], Berengere Husson [b], Lilia Guillet [b], Torstein Pedersen [a]

108 FG

ca 300 species

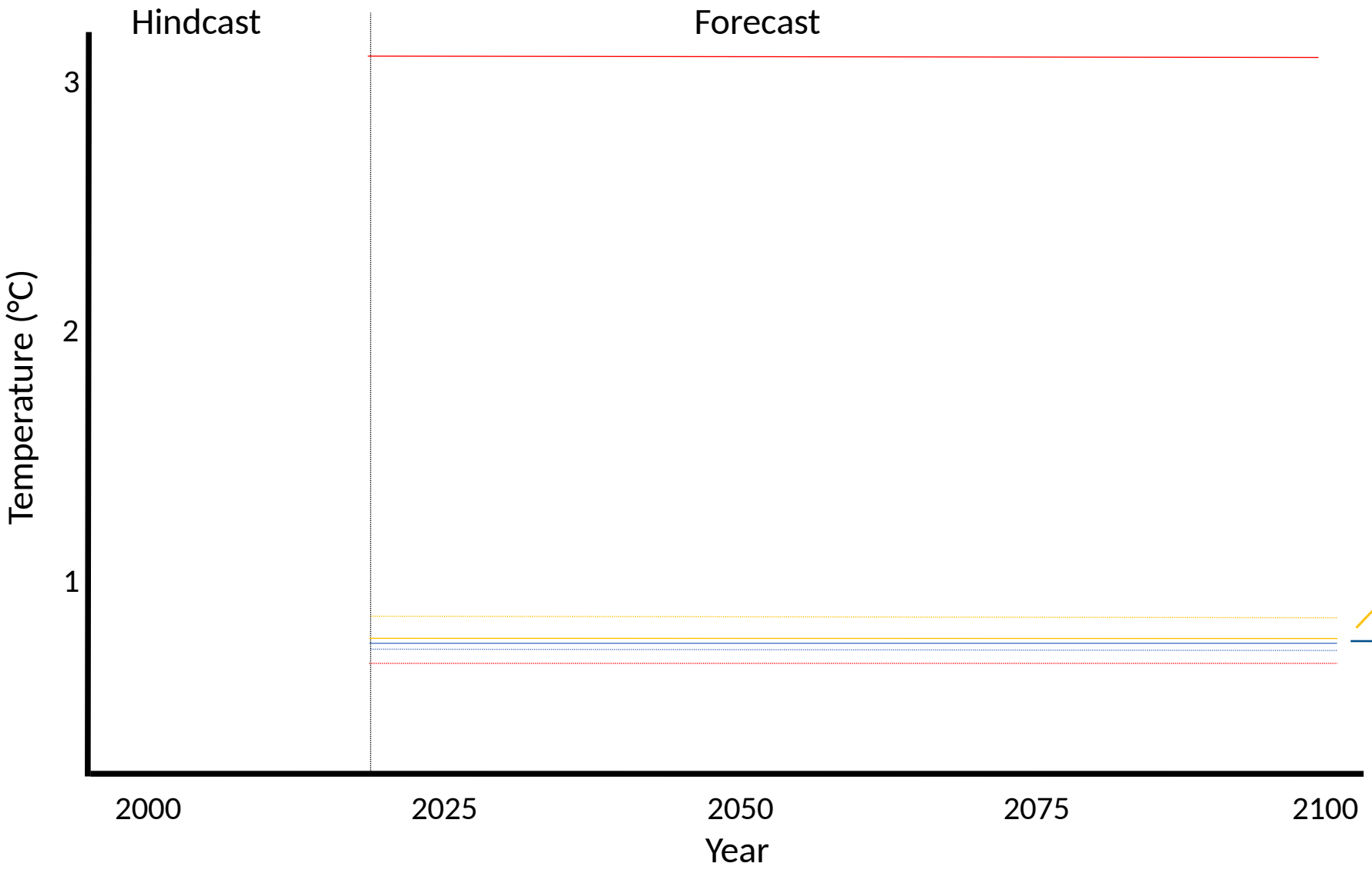
Future scenarios

NEMO NAA 10km

(IMR and UIB)

Surface temperature
Primary production
Sea Ice concentration

Bottom temperature



Change from 2020s to 2090s

+1.65°C
Warm high carbon emission
(SSP 5 - 8.5)

+0.46°C
Intermediate (SSP 2 – 4.5)

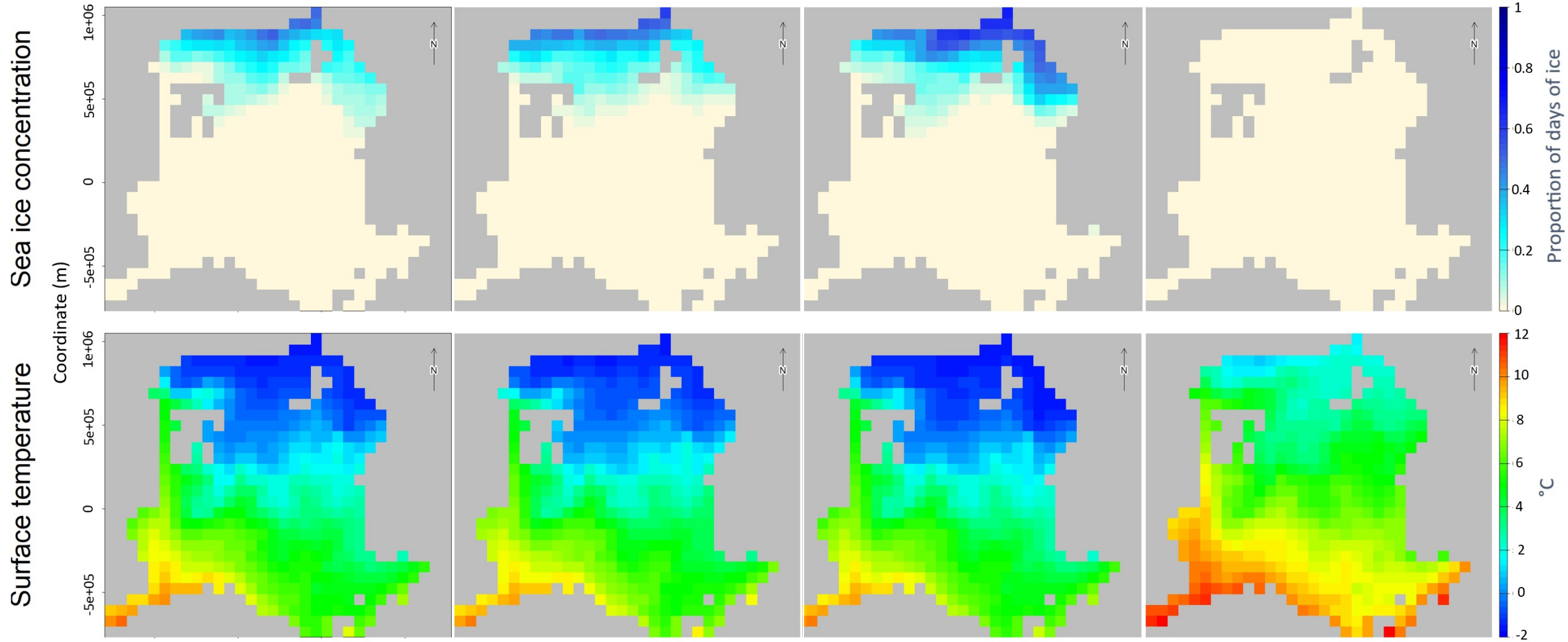
-0.12°C
Cold low carbon emission
(SSP 1 – 2.6)

September

SSP 1 – 2.6

SSP 5 – 8.5

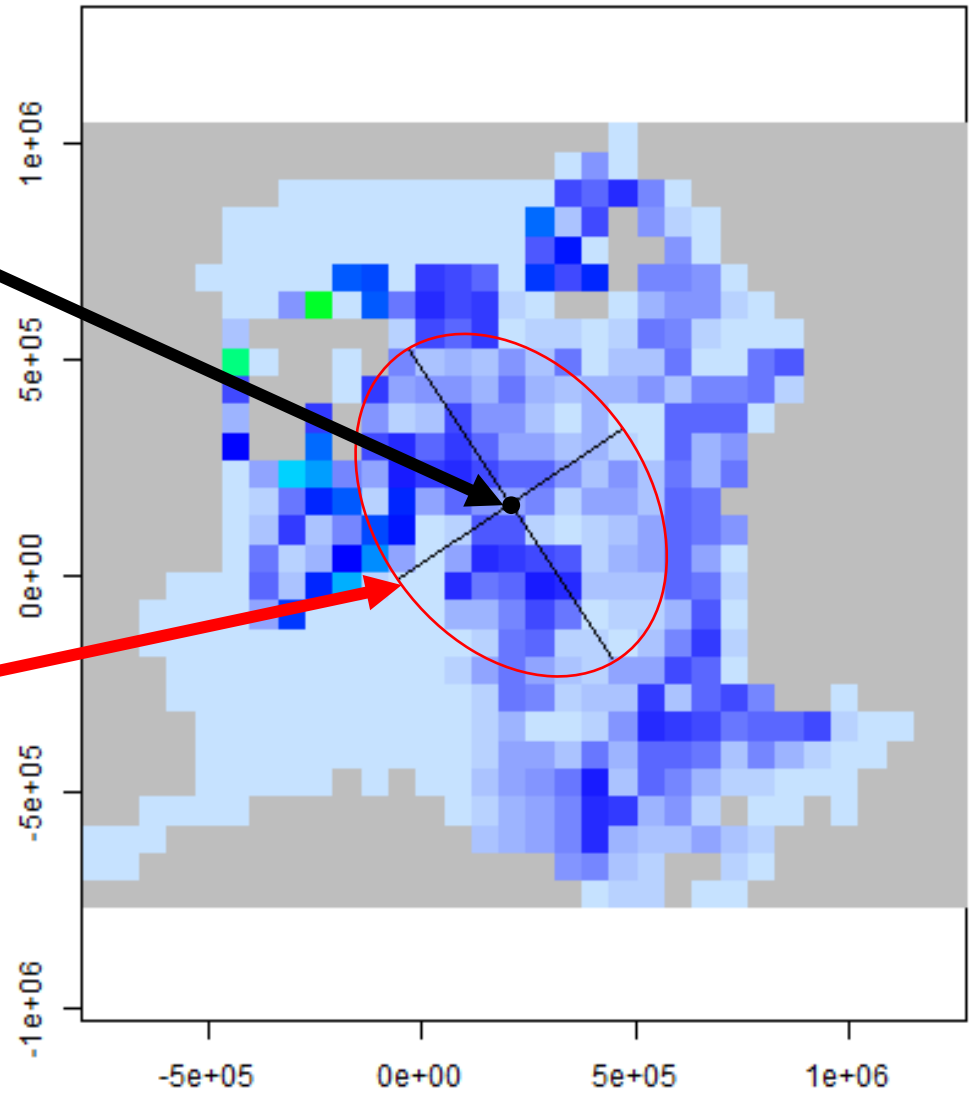
2020’ 2090’ 2020’ 2090’



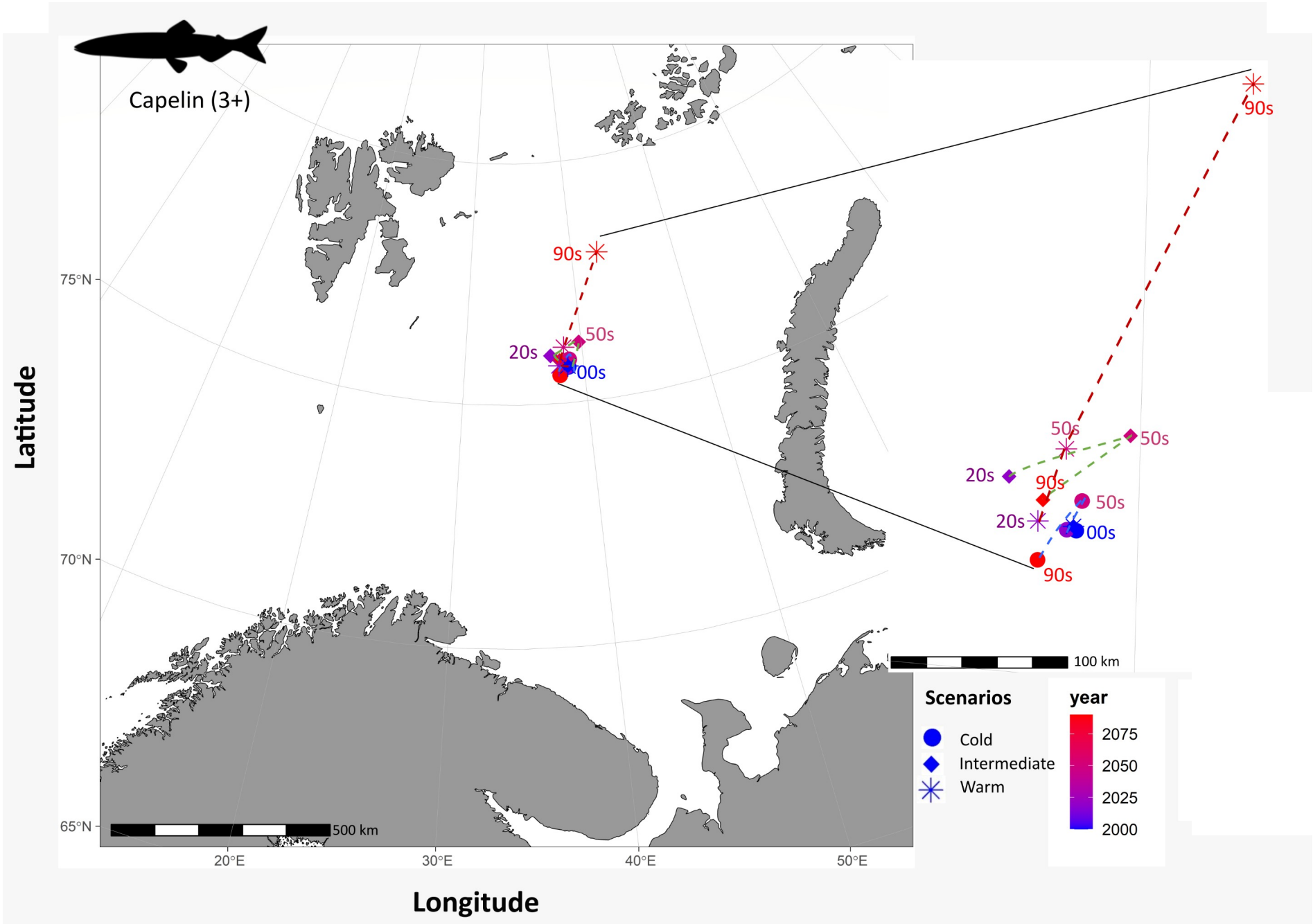
Distribution – relative biomass

Centre of gravity

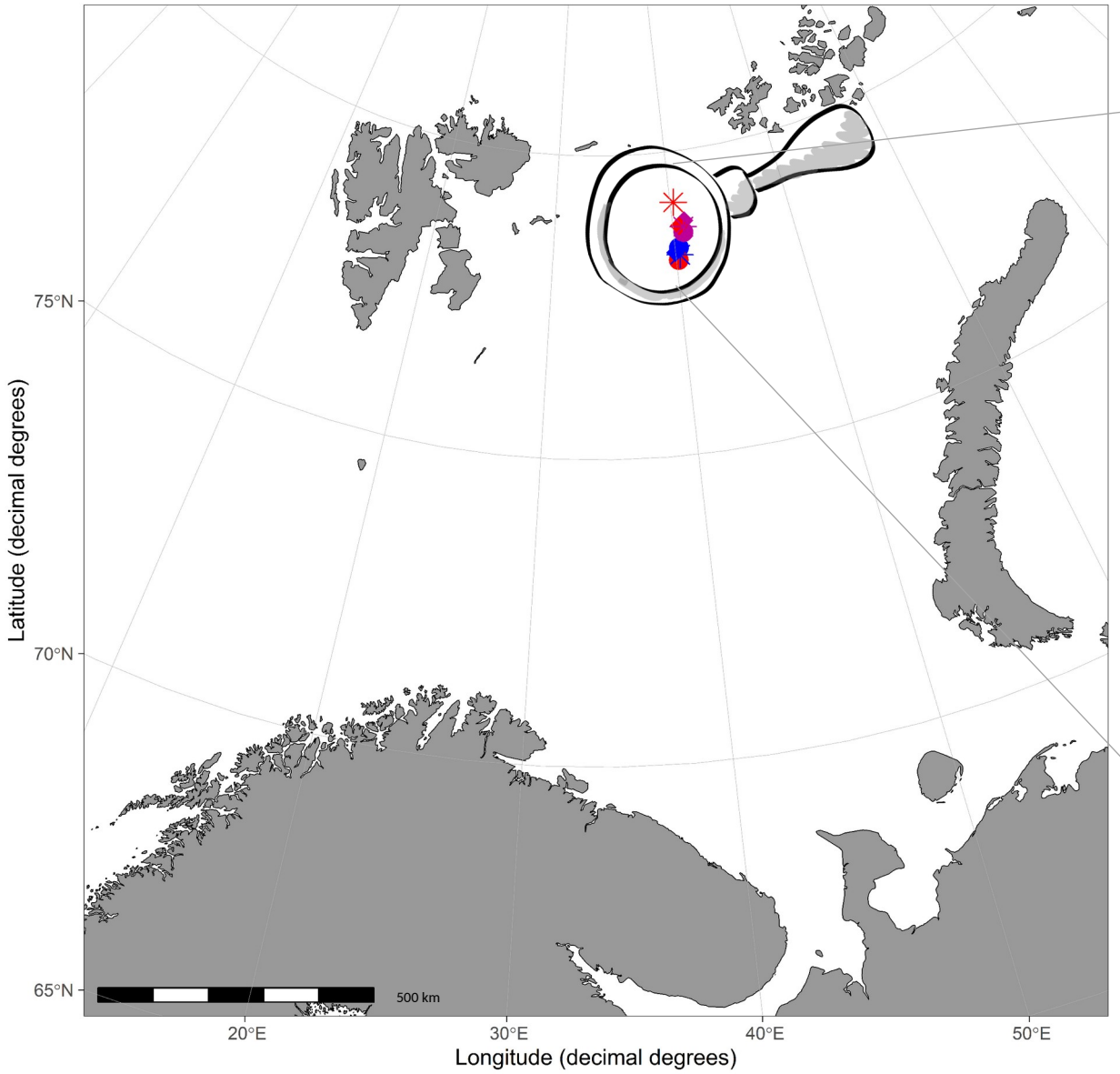
Inertia = spread

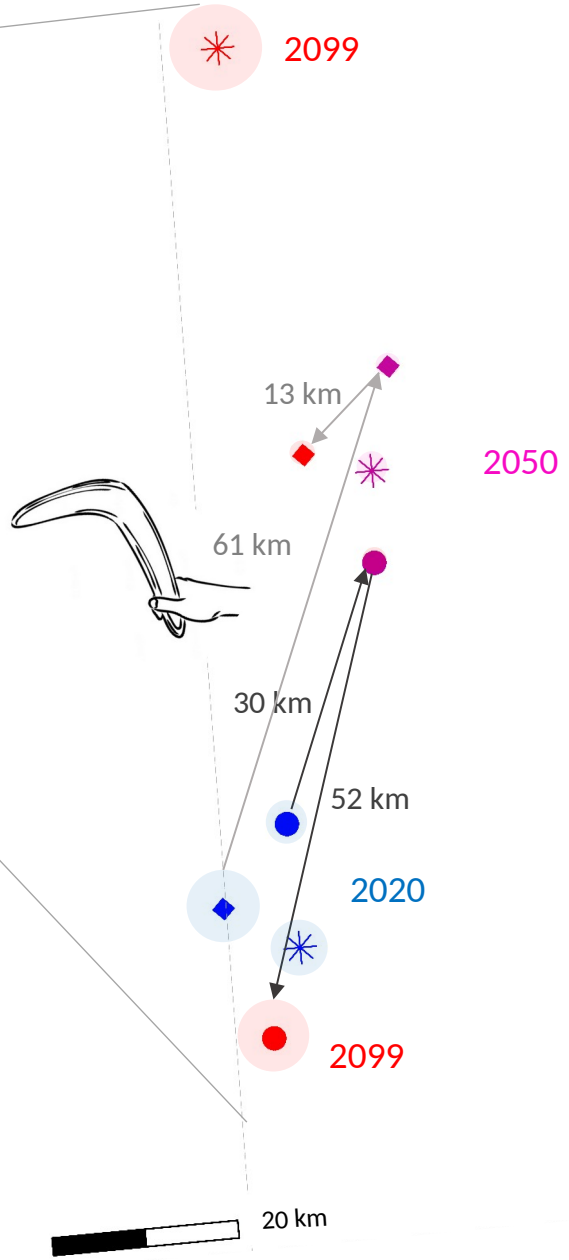


Monthly output for each FG among all simulation


Capelin (3+)
Latitude
Longitude
75°N
70°N
65°N
20°E
30°E
40°E
50°E
90s
50s
20s
00s
500 km
100 km
Scenarios
Cold
Intermediate
Warm
year
2075
2050
2025
2000

Arctic community

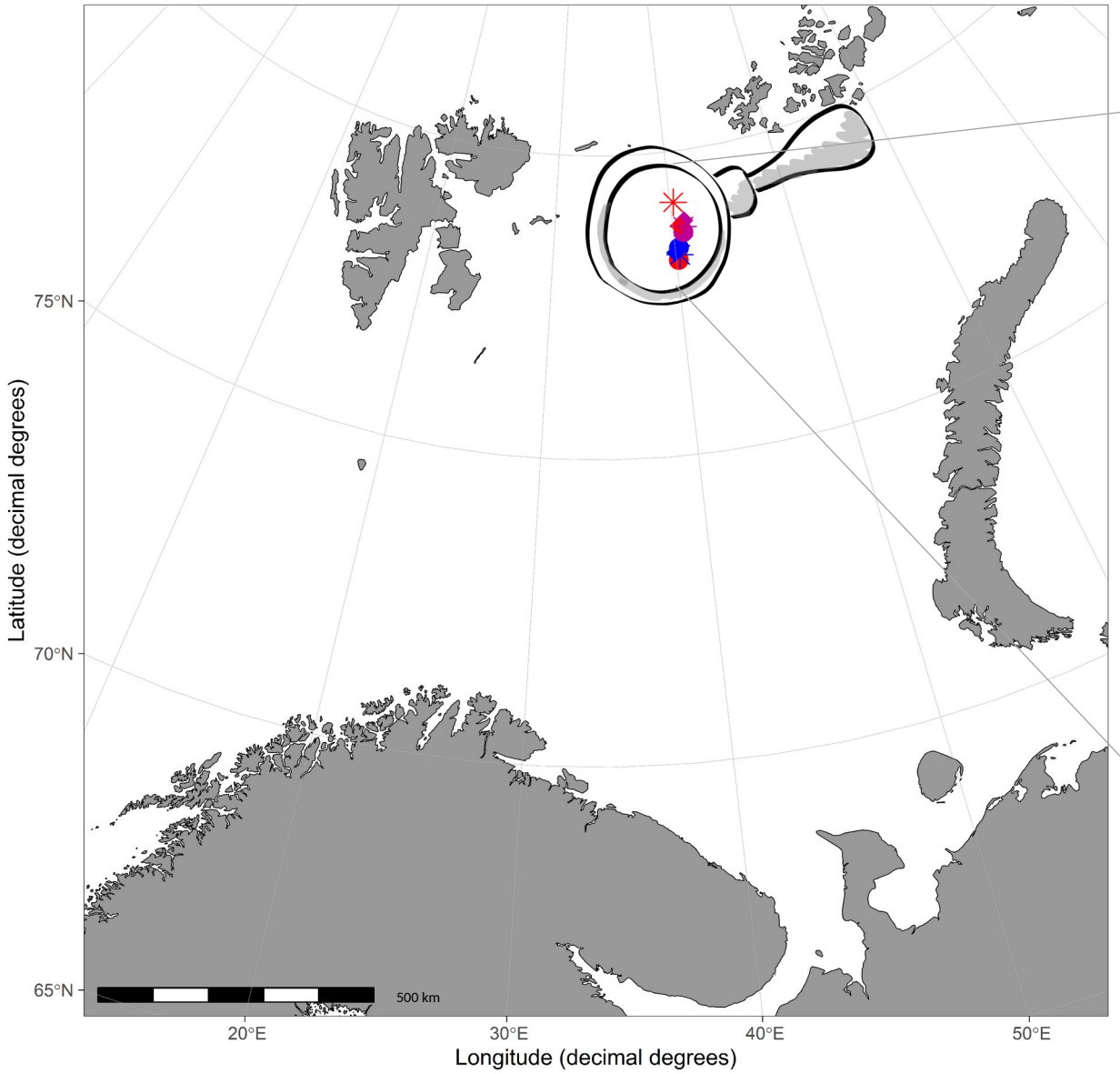


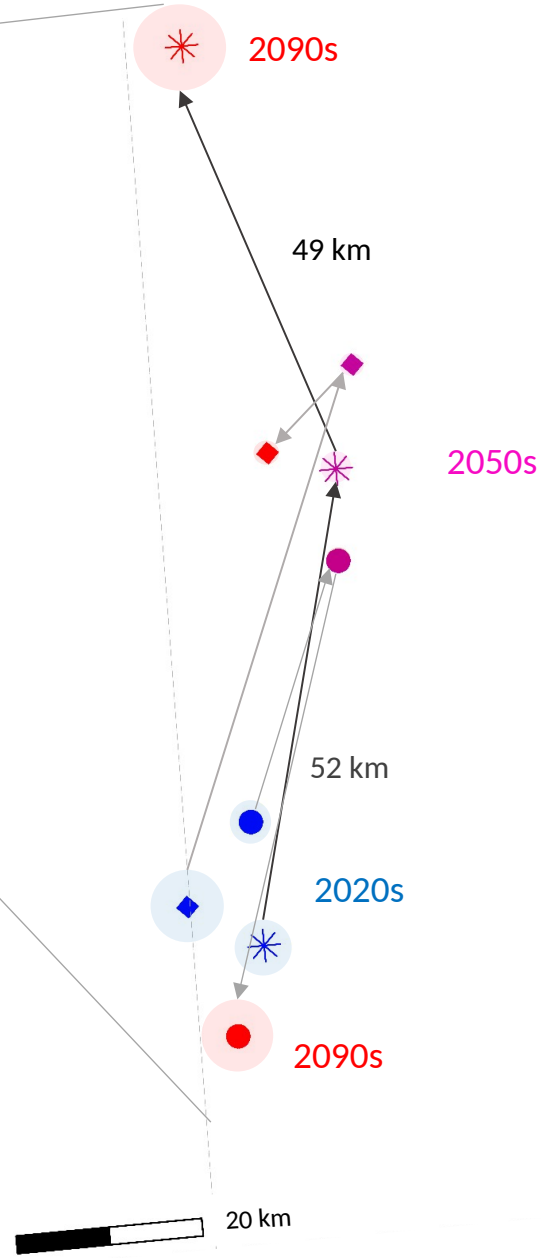


Inertia area
Cold
Intermediate
Warm

year
2080
2060
2040
2020

Arctic community

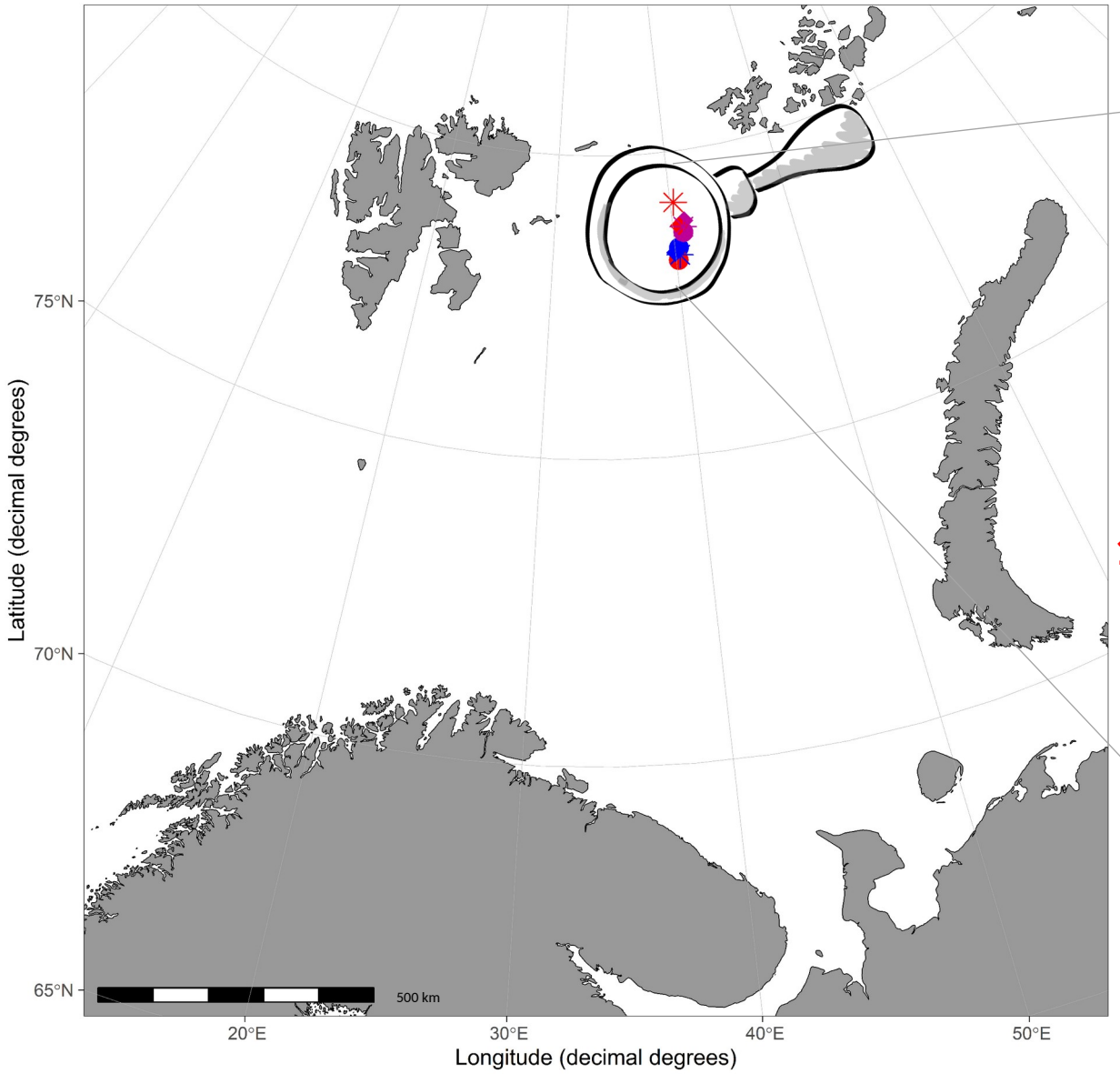


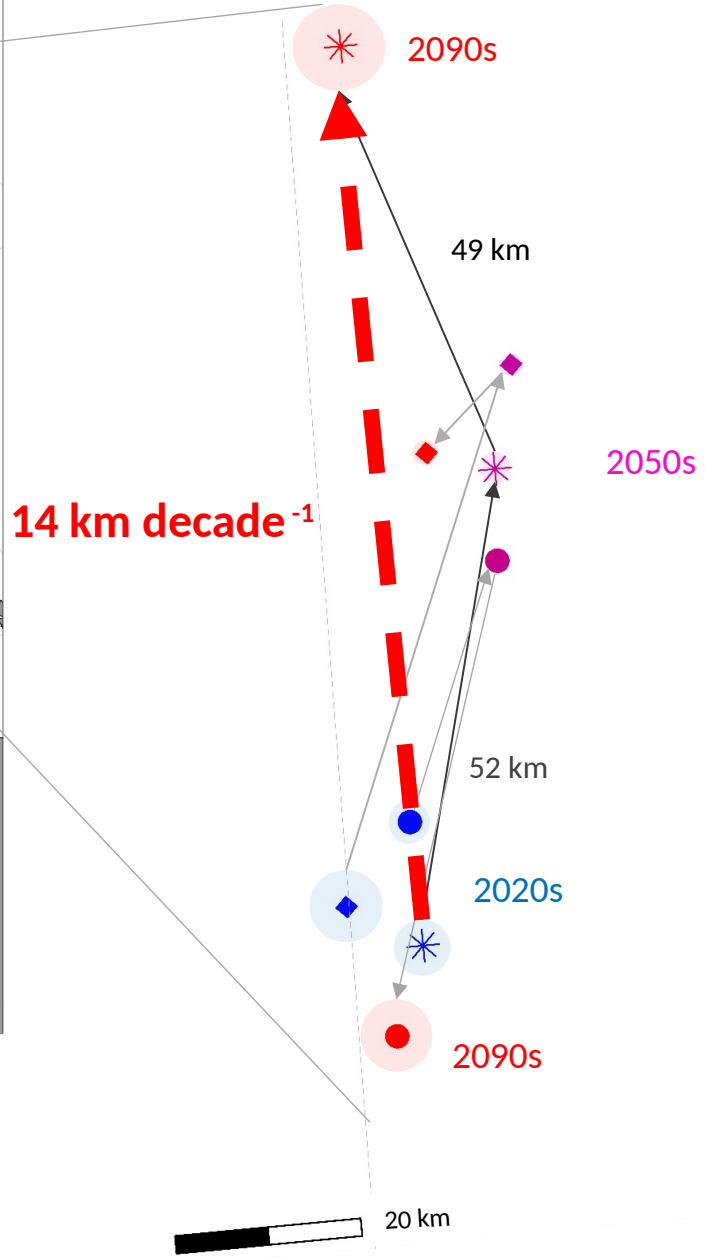


Inertia area
Cold
Intermediate
Warm

year
2080
2060
2040
2020

Arctic community

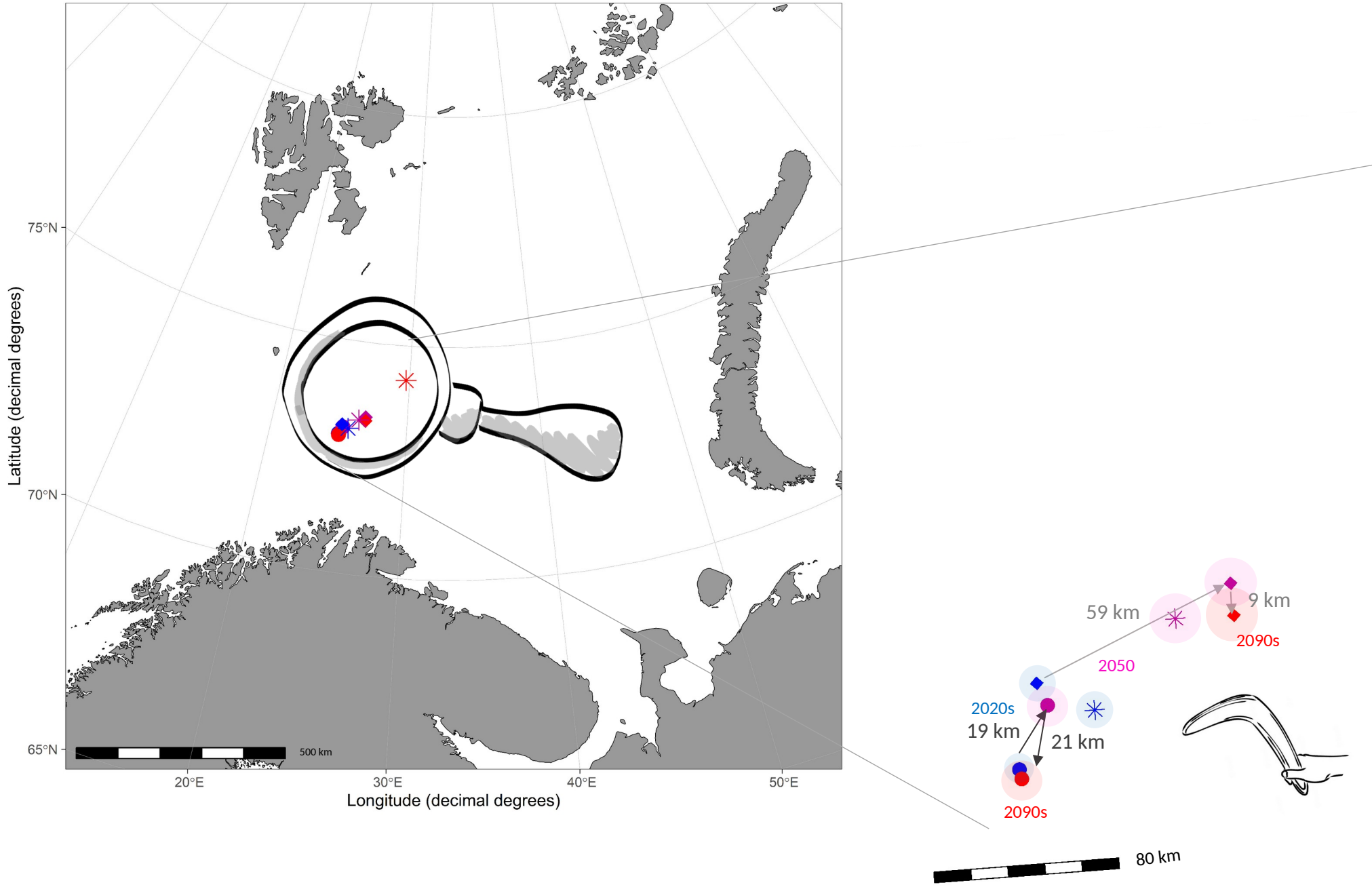




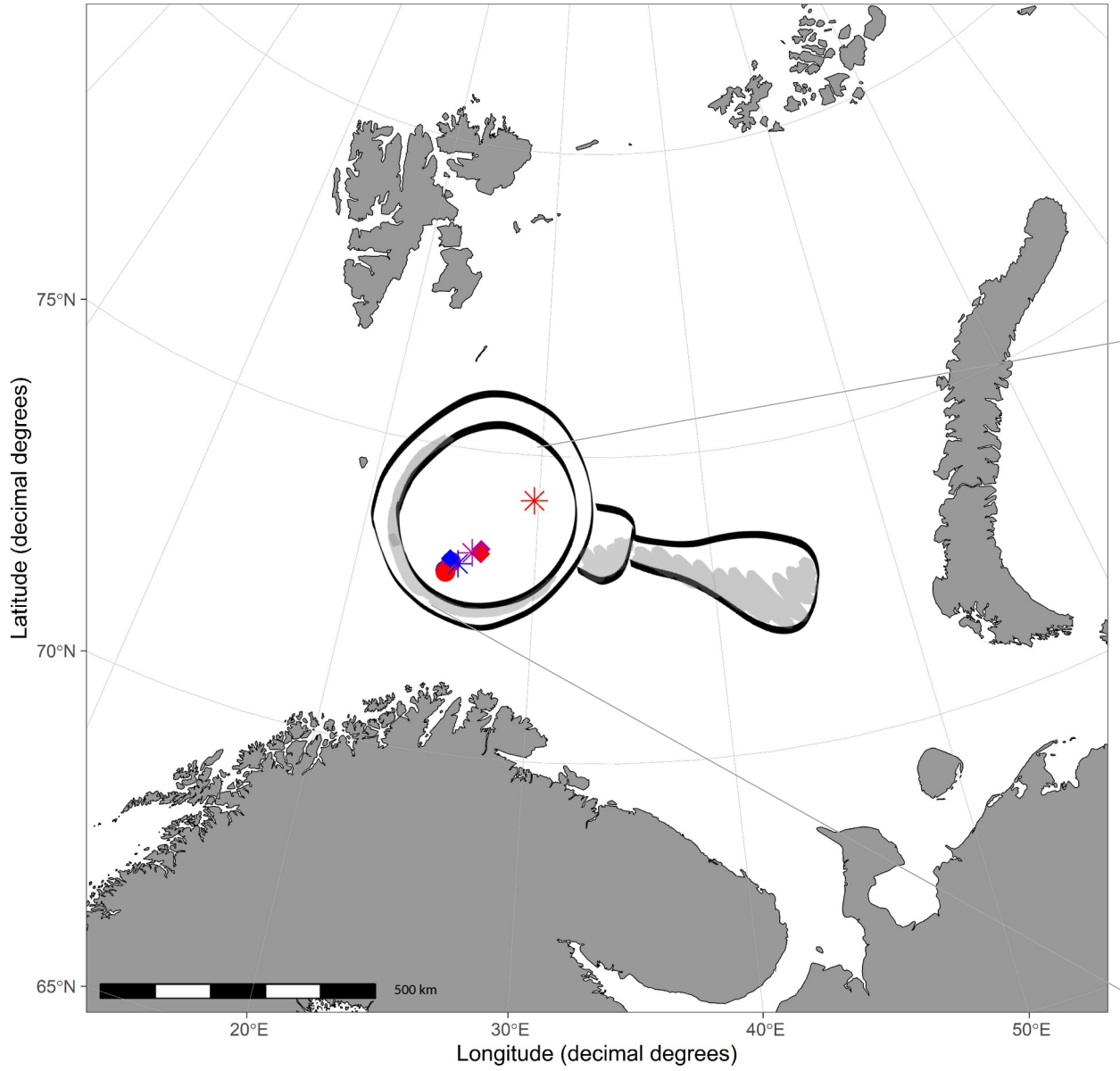
Inertia area
Cold
Intermediate
Warm

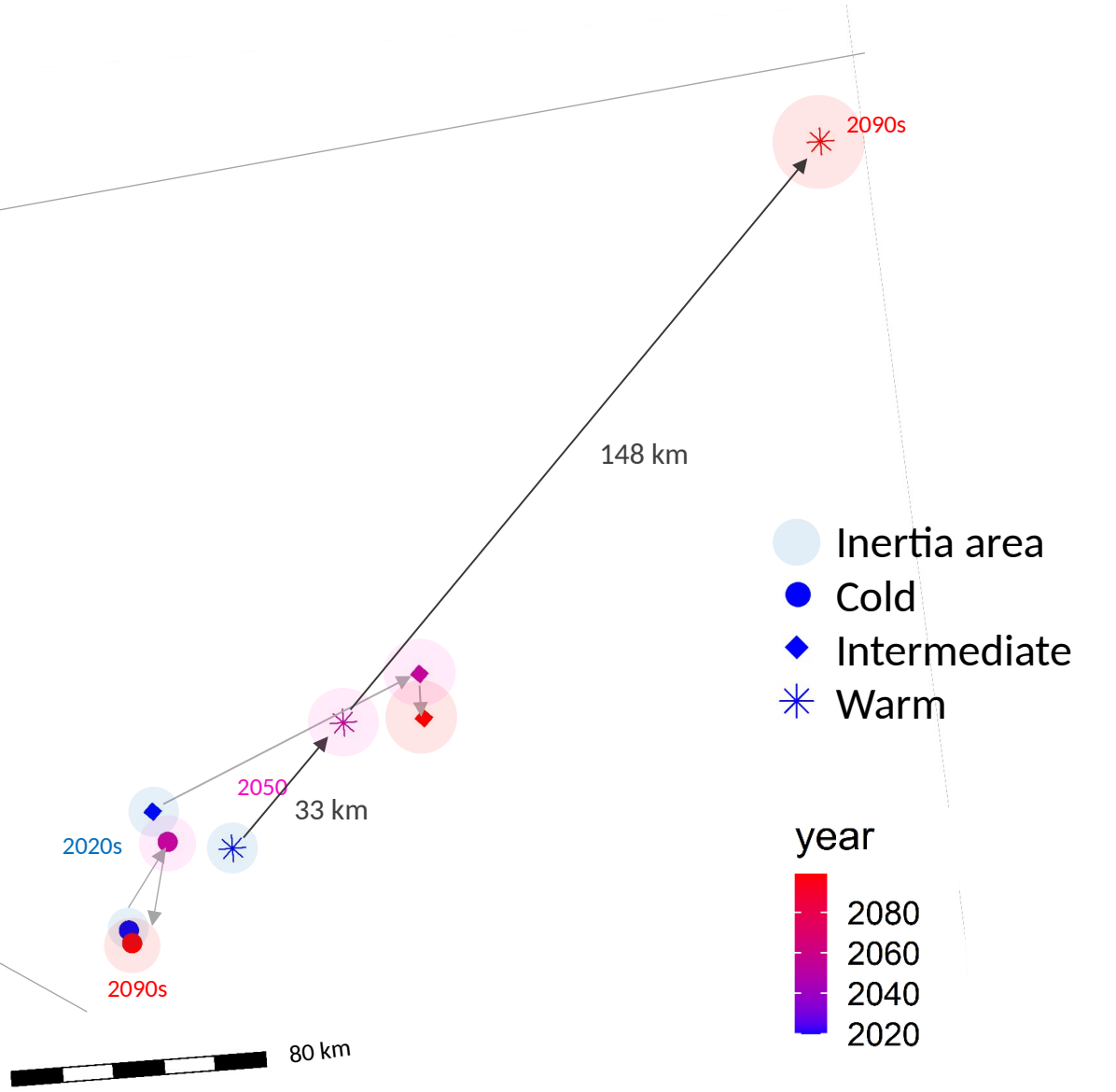
year
2080
2060
2040
2020

Boreal community

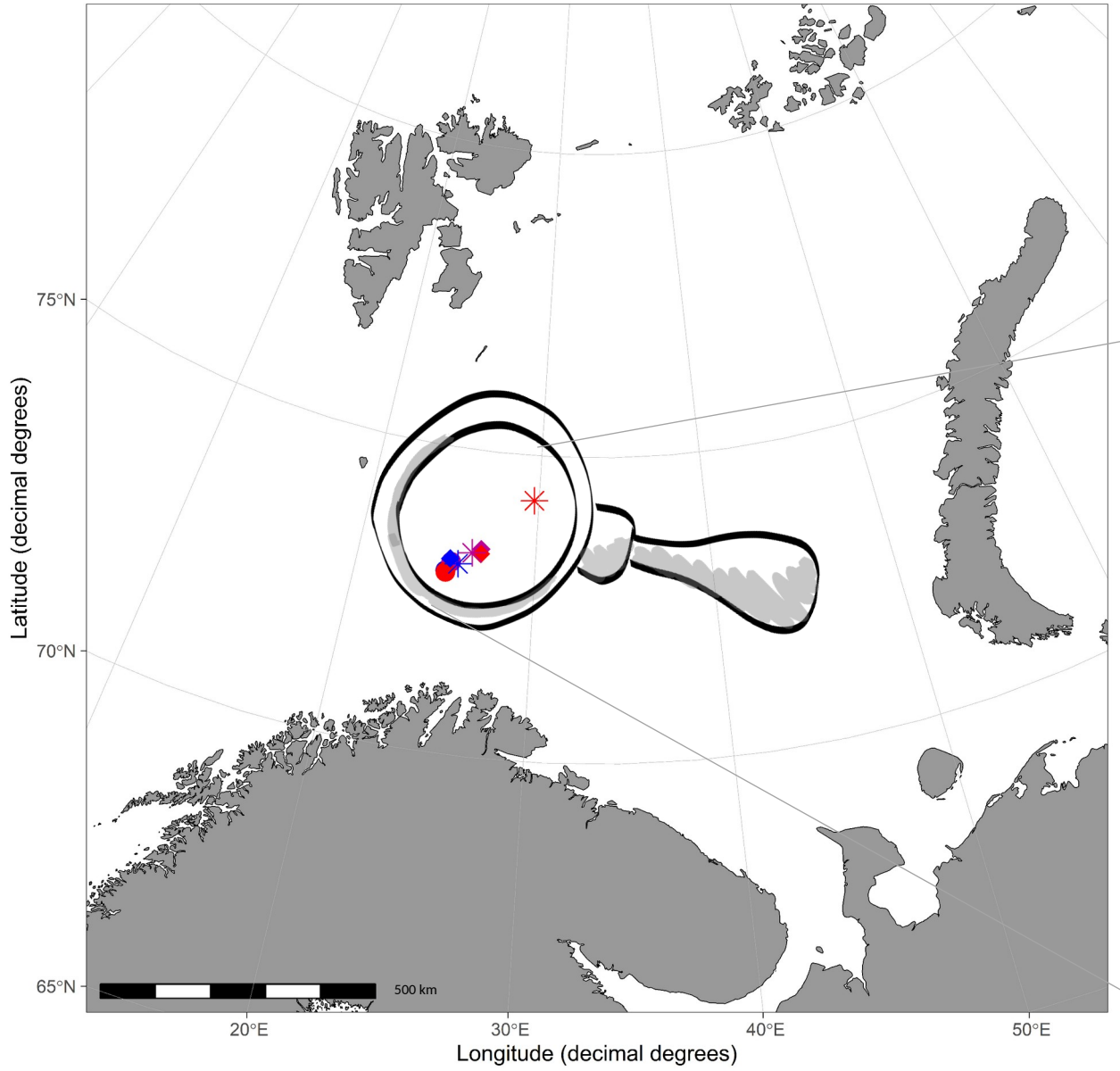


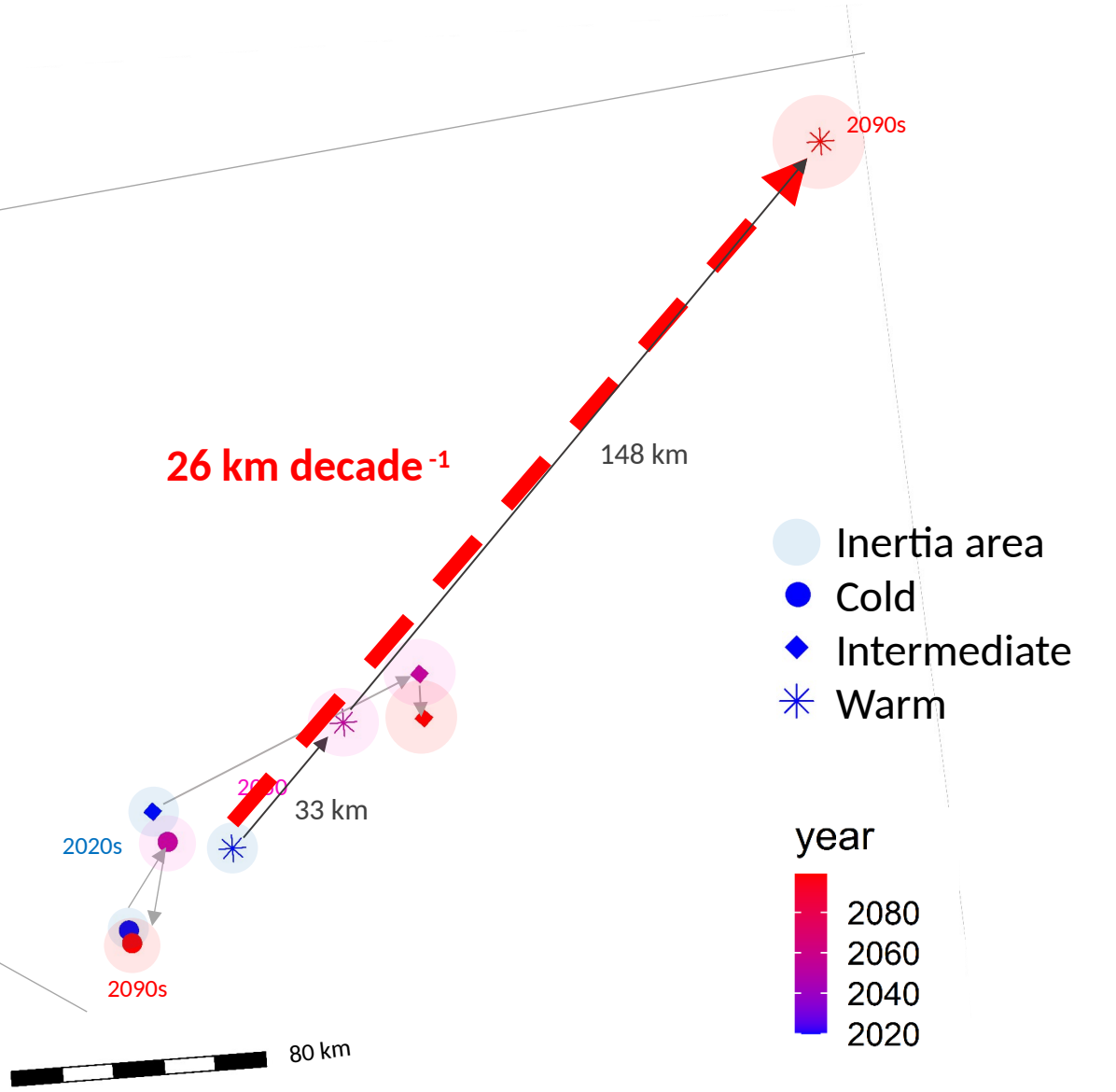
Boreal community





Boreal community





Summarizing

Q1 Will the functional groups (FGs) and the community change the centres and the spread of their spatial distributions at different plausible global warming scenarios?

Yes,

- Cold and intermediate scenarios: FGs shift mainly poleward by 2050 - return to close to 2020's distribution by the end of the century
- Warm: strong poleward shifts in all periods

Summarizing

Q2 Are the responses different in boreal and arctic FGs?

- Yes, arctic FGs shift mainly northward while boreal shift northeast
- Boreal FGs spread more than the Arctic in warm scenarios
- Shift rates are larger in boreal FGs